INCH-POUND
MIL-PRF-9117E
31 December 1996
SUPERSEDING
MIL-A-9117D
19 Apr 1971

PERFORMANCE SPECIFICATION

ADHESIVE; SEALING, FOR AROMATIC FUEL CELLS AND GENERAL REPAIR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a one-part sealing adhesive for fuel cell repair work and for other general repairs where resistance to aromatic fuel is required.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LKJE, 460 2nd Street, STE 221, Robins AFB, GA 31098-1640 by using the Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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2.2.1 Specifications and standards and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-S-735	Standard Test Fluids; Hydrocarbon
FED-STD-313	Material Safety Data, Transportation and Disposal Data for Hazardous Materials Furnished to Government Activities

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, Pennsylvania 19111-5094.)

2.3 Non Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SAE AMS 3217/1A	TEST SLABS, ACRYLONITRILE, BUTADIENE (NBR-H) Medium High Acrylonitrile, 65 - 75

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with paragraph 4.2.

3.2 Material. The material shall be a one-part adhesive suitable for bonding NBR-H surfaces such as those used in aircraft fuel cells.

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3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Bond strength.

3.4.1 Minimum bond strength. The minimum bond strength shall be 5.0 lb. after 4 hours of cure and 10.0 lb. after 24 hours of cure when prepared and tested in accordance with paragraph 4.4.3.2.

3.4.2 Bond strength upon exposure to aromatic fuels. After exposure to aromatic fuels the bond strength shall be a minimum of 6 lb when prepared and tested in accordance with paragraph 4.4.3.3.

3.4.3 Bond strength after storage. The bond strength of cured specimens prepared using aged adhesive shall be at least 80 percent of the strength obtained for the 24 hour cure on the unaged adhesive (see 4.4.4.2).

3.5 Heat resistance. The adhesive, when applied and cured properly, shall be able to withstand exposure to elevated operating temperatures (158°F) without failure of the adhesive bond under a shear load of 2 pounds per square inch (see 4.4.5.2).

3.6 Storage stability. The adhesive, when aged, shall show no signs of gelling, or deterioration such as lumping, coagulation, or separation of adhesive components, and shall be readily brushable (see 4.4.4.1).

3.7 Photochemical requirements. The manufacturer shall certify that materials used in the adhesive are non-photochemically reactive.

3.8 Environmental requirements. The adhesive shall not contain any EPA Class I ozone depleting substances or hazardous materials as defined in FED-STD-313, nor shall it contain any chemical listed in the current report of known carcinogens of the National Toxicology Program.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article testing shall be accomplished using two containers of adhesive (see 3.1). One container shall be used to prepare and test specimens in accordance with paragraphs 4.4.3 and 4.4.5. The second container shall be used for storage stability testing in accordance with paragraph 4.4.4. Failure to pass any examination or test shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance testing shall be accomplished using two containers of adhesive representative of each lot or order. Unless otherwise specified, a lot shall consist of all adhesive manufactured at one time from one batch, forming part of one contract or order, and submitted for inspection at the same time and place. One container shall be used to prepare and test specimens in accordance with paragraphs 4.4.3 and 4.4.5. The second container will be used for storage stability testing, paragraph 4.4.4. Failure of any sample to pass any examination or test shall be cause for rejection of the lot or order.

4.4 Test methods.

Requirements Cross-Reference Matrix

REQUIREMENT	VERIFICATION
3.4.1	4.4.3.2
3.4.2	4.4.3.3
3.4.3	4.4.4.2
3.5	4.4.5.2
3.6	4.4.4.1

4.4.1 Standard conditions. Standard conditions during the make-up of samples and during air cure shall be 77 ± 2°F, and 50 ± 5 % relative humidity.

4.4.2 Test specimens. The standard NBR-H stock used for tests shall conform to AMS 3217/1A or equivalent material as certified by the supplier.

4.4.3 Bond testing. Nine samples will be prepared using naged adhesive as described in paragraph 4.4.3.1. Three will be strip-back tested per paragraph 4.4.3.2 after 4 hours cure time, three will be strip-back tested per paragraph 4.4.3.2 after 24 hours cure time and three will be strip-back tested after immersion in aromatic fuel, paragraph 4.4.3.3.

4.4.3.1 Specimen preparation bond for strength testing. The standard NBR-H stock sheet shall be cut into 1 1/2 inch by 6 inch strips and washed with a suitable solvent. Buffing of the strips will not be allowed. After the strips have dried, apply two coats of adhesive with wait times per manufacturer's recommendations. The adhesive shall be allowed to air cure as specified by the manufacturer, and then reactivated with the manufacturer's recommended solvent. The reactivated surfaces shall be placed together within one minute and appropriate pressure applied in accordance with the manufacturer's

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instructions. After the curing time specified for the particular test, 1 inch by 6 inch strips shall be cut from the center portion of the 1 1/2 inch by 6 inch strips and tested.

4.4.3.2 Strip-back test. Strips bonded in accordance with 3.4.1 to be tested shall be separated by hand at one end of the strip specimen for a distance sufficient to permit the attachment of the jaws of the testing machine. The free ends of the strips shall be clamped in the jaws of a pendulum type test machine and pulled apart at an angle of 180 degrees to each other at a jaw separation rate of 2 inches per minute. The first and last inch of the bonded section shall not be considered in the test. The average constant peak load of each specimen shall be recorded. If any sample falls below the minimum requirement the adhesive shall be rejected.

4.4.3.3 Exposure to aromatic fuels. After preparing three samples as described in paragraph 4.4.3.1 with bond strength as specified in 3.4.2, the samples will be air cured for at least 24 hours at standard conditions before cutting to test size. The samples shall then be fully immersed in standard test fluid conforming to Type III of TT-S-735 for 7 days. After immersion, the strip-back test, 4.4.3.2, shall be conducted. If any sample tested falls below the minimum requirement the adhesive shall be rejected.

4.4.4 Storage stability. One unopened container of adhesive shall be placed in an oven, maintained at a temperature of 120 ± 2° F, and allowed to age 4 weeks. The adhesive shall then be cooled to standard temperature and tested.

4.4.4.1 Visual examination. The adhesive shall conform to the requirements listed in paragraph 3.6 after aging.

4.4.4.2 Bond strength. Contingent upon passing the visual examination of 4.4.4.1, three samples shall be prepared in accordance with paragraph 4.4.3.1 using aged adhesive and tested for bond strength per paragraph 4.4.3.2. Any failure to meet the requirements of paragraph 3.4.3 shall constitute failure of the material.

4.4.5 Heat resistance.

4.4.5.1 Sample preparation. Three specimens shall be prepared for testing in the following manner. Using strips of the standard NBR-H stock, a lap joint shall be made in such a way that the area bonded is 1 square inch. The surfaces shall be cleaned with a suitable solvent prior to application of two coats of the adhesive with wait times per the manufacturer's recommendations. The adhesive will be allowed to air cure as specified by the manufacturer, and then reactivated per the manufacturer's instructions. The bond shall be made within one minute after reactivating, appropriate pressure applied as specified by the manufacturer and air cured at least 18 hours at room temperature before testing.

4.4.5.2 Heat resistance test. The cured samples as specified in 3.5 shall be placed in a dry-air oven and aged for 24 hours at a temperature of 158 ± 2° F. The bond shall then be subjected to a continuous dead-weight shear load of 2 pounds while maintained at the

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elevated temperature for another 24 hours. If any sample fails, the adhesive shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive is intended for use in fuel cell repair work, or for other general repair work, where aromatic fuel resistance is necessary.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual document referenced.
c. Packaging requirements (see 5.1).
d. When a first article is required (see 3.1 and 4.2).
e. Any additional marking requirements (see 6.3).
f, Material Safety Data Sheets, if required (see 6.4).

6.3 Marking. Unless specified otherwise, (see 6.2), containers and boxes shall be marked with appropriate precautionary statements in accordance with the manufacturers standard commercial practice.

6.4 Material Safety Data Sheets. When specified (see 6.2) Material Safety Data Sheets will be provided in accordance with FED-STD-313.

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6.5 Subject term (key word) listing

Test Slabs
Butadiene
NBR-H
Medium HighAcrylonitrile

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99
Army - MR
Navy - AS

Reviewer:
Air Force - 11
Army - AR, MD
Navy - SH, MC

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

Project No.
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-9117E	**2. DOCUMENT DATE (YYMMDD)**

3. DOCUMENT TITLE
Adhesive; Sealing for Aromatic Fuel Cells and General Repair

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**

8. PREPARING ACTIVITY

a. NAME WR-ALC/LKJE	**b. TELEPHONE** *(Include Area Code)* (1) Commercial 912-926-6630 (2) DSN 468-6630
c. ADDRESS ***(Include Zip Code)*** 460 2ND ST STE 221 ROBINS AFB GA 31098-1640	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340 DSN 289-2340**

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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